CST4070 - Weekly challenges

Week 14

> Students are required to solve their challenges at home, submit their solutions to MyLearning (*Assessment — portfolio of data science tasks* folder), and present their solutions in class during the upcoming teaching week.
>
> Successfully completing the weekly challenges will enable students to build their portfolio of tasks, which is essential for earning marks in this module.
>
> For information on how marks are awarded in this module, please refer to the module specifications (*see CST4070 intro*) or consult your teacher.
>
> It's important to note that your instructor will not provide the solutions to any weekly challenges; it is solely the responsibility of students to solve them and propose their solutions in class.

Challenge

Goals

Airbnb is a widely used platform where guests leave reviews about their stays, providing valuable insights into different neighbourhoods. These reviews capture perceptions of safety, affordability, vibrancy, and other qualities, making them a rich source for sentiment analysis and linguistic evaluation. In this challenge, students will leverage Airbnb review data to assess the characteristics of various London neighbourhoods by analysing how closely they align with specific positive and negative descriptors.

Students need to use Airbnb data to answer the following research questions.

Research Questions

1. **Neighbourhood Similarity**. How do different London neighbourhoods compare in terms of key positive and negative descriptors (where the list of descriptors is provided below)?

2. **Sentiment Analysis**. How do neighbourhoods rank based on the overall sentiment of

the Airbnb reviews they receive?

3. **Temporal Stability** (Bonus). Are these sentiment trends stable over time, or do they fluctuate from year to year?

List of Key Descriptors

- **Positive descriptors**: vibrant, charming, peaceful, gastronomic, affordable, historic.

- **Negative descriptors**: isolated, overpriced, noisy, unsafe, crime, gentrified.

Data

You can download the full set of Airbnb listings here and the full set of relative Airbnb reviews here. The key fields contained in your dataset are:

- *Listings*: id, neighbourhood, latitude, longitude
- *Reviews*: listing_id, date, comments

Methodology

To assess the similarity between neighbourhood reviews and predefined descriptors, students can utilise *Word2Vec* or similar word embedding techniques. For sentiment analysis, tools such as *VADER*, *TextBlob*, or *BERT* can be employed to compute sentiment scores for each review.

Deliverable

Participants should submit their Notebook containing the Python code, along with a concise report (in markdown format) outlining their approach, techniques used, and rationale behind their model choices.

Submissions will be evaluated based on the quality of the analysis and the clarity of explanations.

Note. To save your notebook as a PDF file, first export it to HTML, then print it to a PDF file (choose the `Save as PDF` option).